Self-Check Questions: Domain Changes in Algebraic Manipulation

This document was generated by a large language model (LLM) and may contain errors.

Work each question before consulting the answer key at the end. Questions are numbered continuously from 1 to 25; the key uses the same numbers. Where a question asks you to solve an equation, also state *where* any restriction comes from — the original equation's domain, or a step you introduced.

1 Expressions versus functions

1. In what precise sense are $\dfrac{(x+1)\,x}{x}$ and $x+1$ *equal*, and in what sense are they *not*? Name the two settings in which the answers differ.
2. State the natural domain of $\dfrac{x^2-4}{x-2}$. Cancel the common factor; what is the simplified expression and its natural domain? How do the two domains differ?
3. Define "removable singularity." Write the limit that certifies the singularity of $\dfrac{x^2-4}{x-2}$ at $x=2$ is removable, and give its value.
4. True or false, with justification: cancelling a common factor inside a single rational expression can only ever *shrink* its natural domain.
5. Two functions agree at every point of a common set but have different domains. Are they the same function? Explain why the answer flips depending on whether you regard them as functions or as elements of the field $\mathbb{R}(x)$.
6. Construct a rewriting of $y=x+2$ whose natural domain is $\mathbb{R}\setminus\{3\}$ — that is, one that removes exactly the point at $x=3$.

2 Solving equations

7. When you set out to solve an equation, what is the fixed invariant that every manipulation must respect?
8. Distinguish a "blind spot" from an "exclusion." Which of the two, if any, do you carry into the final answer?
9. State, in one sentence, the diagnostic question to ask about any point that appears to be excluded during a solution.
10. For each operation, say whether it can *add* solutions, *drop* solutions, or neither: (a) multiplying both sides by $D(x)$; (b) dividing both sides by $D(x)$; (c) squaring both sides.

11. Explain why "always keep the restriction" is wrong, and why "always discard the restriction" is also wrong. Give the type of example that defeats each rule.
12. When you write "$\dfrac{x^2-1}{x-1} = x+1$ for $x \neq 1$," does the annotation $x \neq 1$ qualify the left-hand side, the right-hand side, or the equality? Explain.
13. True or false, with justification: an extraneous root is always the symptom of an arithmetic mistake.

3 Worked problems

14. Solve $\dfrac{x}{x-3} = \dfrac{3}{x-3}$. Identify any extraneous candidate and trace its restriction to a source.
15. Solve $x^2 = 4x$ two ways: by dividing both sides by x, and by factoring. Which solution does the division lose, and why?
16. Solve $\sqrt{2x+3} = x$. Produce all candidates by squaring, check each in the original equation, and name the extraneous one and the step that produced it.
17. You rewrite the equation $x - 2 = 5$ as $\dfrac{(x-2)(x+4)}{x+4} = 5$ and continue. Does the rewrite change the solution? Which point does it flag, is that point excluded, and what do you do with it?
18. Solve $\dfrac{1}{x-1} + \dfrac{1}{x+1} = \dfrac{2}{x^2-1}$. State the solution set and explain it in terms of where the restriction originates.
19. For $\dfrac{x^2-x}{x} = 1$, answer in order: (a) natural domain of the left side; (b) its simplification; (c) the candidate solution; (d) whether any candidate is excluded; (e) the final solution set.

4 Diagnosing fallacies

20. In the derivation $a = b \Rightarrow a^2 = ab \Rightarrow a^2 - b^2 = ab - b^2 \Rightarrow (a+b)(a-b) = b(a-b) \Rightarrow a + b = b \Rightarrow 2 = 1$, identify the illegal step and the exact value of the quantity being divided by.
21. Diagnose the fallacy $-1 = \sqrt{-1} \cdot \sqrt{-1} = \sqrt{(-1)(-1)} = \sqrt{1} = 1$. Which identity is misapplied, and what is its domain of validity?
22. A "proof" starts from $x = -3$, squares to get $x^2 = 9$, and concludes $x = 3$. Explain the error using the taxonomy of risky steps.
23. Construct your own fallacious argument that $0 = 3$ by concealing a division by zero. State plainly where the division by zero occurs.

5 Synthesis

24. Explain why, in a typical calculus exercise, you can divide by expressions repeatedly and almost never be burned. What conditions make a blind spot harmless?

25. Give one example of a domain restriction that is part of a *function's identity*, and one example of a restriction that is merely an *artifact of a solving step*. State clearly which is which.

Answer key

1. They are equal as *formal rational expressions* — the same element of the field $\mathbb{R}(x)$, where cancelling a common factor is legal. They are *not* equal as *functions*: the first has natural domain $\mathbb{R} \setminus \{0\}$, the second has domain $\mathbb{R}$, and a function includes its domain.
2. Natural domain: $x \neq 2$. Cancelling, $\dfrac{(x-2)(x+2)}{x-2} = x + 2$ for $x \neq 2$. The simplified form $x + 2$ has domain $\mathbb{R}$. The two differ only at $x = 2$, where the original has a hole that the simplified form fills.
3. A removable singularity is a point absent from the natural domain at which the two-sided limit exists (and is finite), so the function can be patched to be continuous there. Here $\lim_{x\to 2} \dfrac{x^2-4}{x-2} = \lim_{x\to 2}(x+2) = 4$.
4. False. Cancelling a common factor *enlarges* the domain (it fills a hole). It is *introducing* a denominator that shrinks the domain.
5. As functions: no, they are not the same, because the domain is part of a function's identity. As elements of $\mathbb{R}(x)$: yes, they are the same object, because that field identifies fractions equivalent under cross-multiplication and never evaluates at a point. Function equality demands equal domains and equal values; field equality demands only the algebraic equivalence.
6. For instance $y = \dfrac{(x+2)(x-3)}{x-3}$. The factor $\dfrac{x-3}{x-3}$ equals 1 except at $x = 3$, where it is $0/0$; this removes exactly the point $x = 3$.
7. The solution set of the *original* equation. It is fixed before any manipulation; each step is only a means of reading it.
8. A *blind spot* is a point where a step is undefined or fails to preserve equivalence, so the method yields no information there; you must test it in the original equation. An *exclusion* is a point genuinely outside the original equation's domain; it cannot be a solution. You keep an exclusion (as forbidden); you never simply keep a blind spot — you check it.
9. "Is this point excluded by the original equation itself, or only by a step I introduced?"
10. (a) Multiplying by $D(x)$ can *add* solutions (at the zeros of D) but cannot drop any. (b) Dividing by $D(x)$ can *drop* solutions (at the zeros of D) but cannot add any. (c) Squaring can *add* solutions (it is not injective) but cannot drop any.
11. "Always keep" fails when a restriction is step-induced and the excluded point is in fact a solution — e.g. rewriting $x + 1 = 1$ as $\dfrac{x(x+1)}{x} = 1$ would wrongly report no solution, though $x = 0$ solves the original. "Always discard" fails when the restriction belongs to the original equation's domain — e.g. keeping $x = 1$ as a solution of an equation whose denominators are $x - 1$ admits a value where the equation is undefined.
12. It qualifies the *left-hand side* (and therefore the equality). The right side $x + 1$ is perfectly defined at $x = 1$; the equation holds only where the left side is defined, namely $x \neq 1$.
13. False. An extraneous root is the *expected* byproduct of a step that does not preserve equivalence (multiplying by a vanishing factor, squaring). The error would be failing to check candidates and discard it, not its appearance.
14. Original domain: $x \neq 3$. Multiplying by $x - 3$ gives $x = 3$, which is excluded by the original domain. The restriction comes from the *original* equation, so $x = 3$ is a genuine exclusion: it is extraneous and rejected. Solution set: empty.

15. Factoring: $x^2 - 4x = 0 \Rightarrow x(x-4) = 0 \Rightarrow x \in \{0, 4\}$. Dividing by x gives only $x = 4$, losing $x = 0$, because the division silently assumed $x \neq 0$ while $x = 0$ was a genuine solution.
16. Squaring: $2x + 3 = x^2 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x-3)(x+1) = 0 \Rightarrow x \in \{3, -1\}$. Check: $\sqrt{9} = 3$ holds; $\sqrt{1} = 1 \neq -1$ fails. The root $x = -1$ is extraneous, introduced by the (non-injective) squaring step. Solution: $x = 3$.
17. The rewrite does not change the solution: $x = 7$. It flags $x = -4$, a *step-induced* point, so it is a blind spot, not an exclusion. Checking it in the original: $-4 - 2 = -6 \neq 5$, so $x = -4$ is not a solution anyway — but the rule is that had it satisfied the original, you would keep it. Solution set: $\{7\}$.
18. Original domain: $x \neq \pm 1$. The left side combines to $\dfrac{(x+1)+(x-1)}{x^2-1} = \dfrac{2x}{x^2-1}$, so the equation becomes $2x = 2$, giving $x = 1$. But $x = 1$ is excluded by the original domain, so it is extraneous. Solution set: empty.
19. (a) $x \neq 0$. (b) $\dfrac{x(x-1)}{x} = x - 1$ for $x \neq 0$. (c) $x - 1 = 1 \Rightarrow x = 2$. (d) $x = 2$ is not excluded. (e) Solution set: $\{2\}$.
20. The illegal step is dividing both sides by $a - b$. Because $a = b$, the quantity $a - b$ is exactly 0, so this is a division by zero.
21. The misapplied identity is $\sqrt{ab} = \sqrt{a}\,\sqrt{b}$, which is valid only for $a, b \geq 0$ (nonnegative reals). With $a = b = -1$ it fails, so the second equality is invalid.
22. Squaring is not injective: both $x = 3$ and $x = -3$ satisfy $x^2 = 9$, so squaring *added* the candidate $x = 3$. Concluding $x = 3$ keeps the extraneous root and discards the true value $x = -3$; checking against the original $x = -3$ would have caught it.
23. One construction: for any number a, both $a \cdot 0 = 0$ and $(a+3) \cdot 0 = 0$, so $a \cdot 0 = (a+3) \cdot 0$. "Cancelling the common factor 0" gives $a = a + 3$, hence $0 = 3$. The division by zero is the cancellation of the factor 0.
24. Because the expressions you divide by are generically nonzero on the domain of interest, the values that would trigger a blind spot usually are not solutions, and the extraneous-root check (when performed) passes trivially. A blind spot is harmless precisely when the divisor is provably nonzero on the relevant domain, or when you check the flagged candidates regardless.
25. Function-identity restriction: $f(x) = \dfrac{1}{x}$ has domain $x \neq 0$ permanently — it is part of what f *is*. Solving-step artifact: rewriting the equation $x + 1 = 1$ as $\dfrac{x(x+1)}{x} = 1$ introduces a temporary $x \neq 0$ that belongs to the detour, not to the problem.